



PRODUCT REQUIREMENTS (PRD)

# Tyre DMS & Inventory Ecosystem

Mobile App Product Requirements Document & UX Specification

## 1. Executive Summary

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Tyre retail and B2B distribution operations carry specialized complexities compared to general inventory retail. Products are physically heavy, take up substantial storage space, and feature highly specific, safety-critical parameters. Furthermore, dealers maintain low on-hand stock and rely heavily on quick-turnaround local distributor delivery when customized vehicle requests arrive.

This Product Requirements Document (PRD) describes an integrated, mobile-first ecosystem consolidating inventory tracking, automated par reorders, real-time B2B quotes, and discrepancy audits.

## 2. Specialized Tyre Taxonomy (Core Data Model)

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Standard retail schemas fail to capture tyre specifications. All cataloging and search modules must support the following structure:

| Attribute               | Type / Format    | Description                             | Example                    |
|-------------------------|------------------|-----------------------------------------|----------------------------|
| <b>Brand</b>            | Enum / String    | Tyre manufacturer                       | Apollo , CEAT , MRF        |
| <b>Pattern / Model</b>  | String           | Tread pattern and branding identifier   | Milaze X3 , SecuraDrive    |
| <b>Section Width</b>    | Integer (mm)     | Tread width from sidewall to sidewall   | 165 , 205 , 185            |
| <b>Aspect Ratio</b>     | Integer (%)      | Sidewall height relative to tread width | 70 , 60 , 65               |
| <b>Rim Diameter</b>     | Integer (Inches) | Wheel diameter size fit                 | 14 , 16 , 15               |
| <b>Vehicle Category</b> | Enum             | Segment fitment                         | Four Wheeler , Two Wheeler |
| <b>Condition</b>        | Enum             | Tyre condition status                   | New , Used , Retread       |
| <b>DOT Code</b>         | String (WWYY)    | Manufacturing week and year             | 1224 (12th week of 2024)   |

### 3. B2B Quote Bidding Marketplace

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To resolve dealer shortages, the app features an active RFQ (Request for Quote) marketplace allowing local suppliers to bid directly on order requests.

#### B2B Bidding Flow Sequence

1. Dealer -> Submits RFQ (Size, Qty, Timelimit)
2. System -> Broadcasts to local Suppliers
3. Suppliers -> Submit Price & Time Bids
4. System -> Calculates Dealer Profit Margins
5. Dealer -> Accepts Bid via BNPL/Credit limit
6. System -> Generates Purchase Order (PO)

#### Core Features:

##### 1. Live Timers

Quotes remain active for limited durations (e.g. 3h 49m) to drive immediate supplier actions and secure order placement.

##### 2. Profit Margin Estimator

Automatically calculates potential profit margin based on landed costs vs. suggested retail prices before bid confirmation.

##### 3. BNPL Split Payments

Seamlessly splits transactions between cash, credit limits, and credit note (CN) deductions.

##### 4. Supplier Notifications

Instant SMS/Push notifications alert suppliers to submit competing bids for regional stock requests.

### 4. Specialized Inventory Intake

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- **OCR Invoice Scans:** Snap invoice photos to automatically parse SKU details, buying rates, and quantities directly into stock lists.
- **Multi-Branch Sync:** Query sister store warehouses for sizes prior to submitting supplier quote requests.
- **Auto-Reorder Triggers:** Par level trackers automatically create draft POs once core hatchback or two-wheeler tyre quantities fall below the threshold.

## 5. Mobile-First Stocktake reconciliation

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Designed to address challenging warehouse environments with weak network connectivity and low-light parameters.

### Audit Process Rules

1. **Double Audit Method:** Camera scanner counts combined with large tactile increment buttons (+/-) for manual entries.
2. **Visual Delta Signals:** Shortages are colored in Brand Red ( #ED1D24 ); matches are shown in Success Green ( #00B633 ).
3. **Offline Cache Queue:** Stock counts are saved locally and synchronized automatically when Wi-Fi becomes available.
4. **Manager Verification:** Variances exceeding 2 items or values of ₹10,000 require manager approval and a secondary verification count.

## 6. Design System & UI Guidelines

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Interfaces must follow a clean, accessible layout:

- **Colors:** Primary Brand Red ( #ED1D24 ) for highlights and primary CTAs, Success Green ( #00B633 ) for verified counts, and Slate Gray ( #627085 ) for details.
- **Typography:** Poppins for UI, buttons, and titles; Inter / SF Pro for technical text, section dimensions, and timestamps.
- **Dimensions:** 16px layouts on mobile margins, with corner radii values of 4px for badges, 12px for buttons, and 24px for top sheet corners.